FINITE ALGEBRA DATA FOR SUPERCUSPIDAL REPRESENTATIONS

Abstract. We give finite algebra data for irreducible smooth complex supercuspidal representations of a connected reductive group over a nonarchimedean local field. At a fixed compact open level and central character, nilpotence relations for positive Cartan operators define finite-dimensional semisimple Hecke quotients. A finite boundary calculation determines their multiplication tables. Their simple modules reconstruct the supercuspidal representations without an additional choice of an irreducible quotient. A rational certificate for the classical finiteness of Hecke algebras over their centers gives a terminating search for a uniform stopping bound. We state the parameters as finite matrices, give their equivalence relation, and describe the associated group representation by generators and relations. The construction has no restriction on residue characteristic or splitting of the group. Algorithmic statements require specified effective local group operations and exact coefficient arithmetic. The construction generally gives a quotient of a compact induction; in the sector with nilpotence exponent one it gives a single irreducible compact induction from a vertex stabilizer.

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1. The result and the data

Let F be a nonarchimedean local field, let $\mathbf{G}$ be a connected reductive F-group, and put $G = \mathbf{G}(F)$ and $Z = Z(G)$. All representations in this article are smooth complex representations. We fix a decreasing sequence $(D_e)_{e\geq 0}$ of compact open subgroups at one special vertex, as specified in Section 2. The sequence is a neighborhood basis of the identity. The central character will always mean an actual character $\omega : Z \to \mathbf{C}^\times$, acting by scalars, rather than a generalized central character.

The result gives data for the supercuspidals with D_e-fixed vectors and central character ω. The data are simple modules of one finite algebra $Q^{\mathrm{sc}}_{e,\omega}$. This algebra is obtained from double-coset convolution by imposing powers of explicitly chosen positive Cartan operators. Its construction does not require a list of supercuspidal representations.

Theorem 1.1. *For each e and each smooth ω trivial on $Z \cap D_e$, the construction in Sections 2–5 gives a finite-dimensional semisimple complex algebra $Q^{\mathrm{sc}}_{e,\omega}$ with the following properties.*

(i) Each simple module L determines an irreducible supercuspidal representation $\Pi_{e,\omega}(L)$ with $\Pi_{e,\omega}(L)^{D_e} \simeq L$ and central character ω.

(ii) Every irreducible supercuspidal representation with these fixed vectors and central character occurs, and two simple modules give isomorphic group representations exactly when they are isomorphic.

(iii) Under the effective input conventions of Section 7, there is a terminating procedure to compute the multiplication table of $Q^{\mathrm{sc}}_{e,\omega}$ and its simple modules.

There is no restriction on the residue characteristic or on the splitting field of $\mathbf{G}$.

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Here is the corresponding matrix form of the data. Suppose the computed algebra is nonzero, and choose a basis $q_1 = 1, q_2, \dots, q_n$. Write $q_i q_j = \sum_k c_{ij}^k q_k$. A datum at (e, ω) is a positive integer d and matrices $X_1, \dots, X_n \in M_d(\mathbf{C})$ satisfying

$$X_1 = I_d, \qquad X_i X_j = \sum_k c_{ij}^k X_k, \qquad \operatorname{span}_{\mathbf{C}}\{X_1, \dots, X_n\} = M_d(\mathbf{C}). \tag{1.1}$$

The last condition is a finite rank test and is equivalent to simplicity of the module. Two such data are identified by simultaneous conjugation. If the algebra is zero, there are no data at that level and central character. Section 6 gives the reconstruction explicitly as a space of symbols with specified relations and group action.

To parametrize all supercuspidals without repetitions across levels, retain only data for which averaging over D_{e-1} is zero when $e > 0$. Equivalently, e must be the least level with nonzero fixed vectors. The central character, least level, and simultaneous-conjugacy class of the matrices are then complete parameters. One can replace the matrix class by a primitive central idempotent of $Q_{e,\omega}^{\mathrm{sc}}$.

The proof has three parts. Positive Cartan nilpotence produces a finite quotient and forces Jacquet modules to vanish. Projectivity of supercuspidals with fixed central character then makes this quotient semisimple and makes reconstruction irreducible. Finally, a finite rational certificate for finiteness over the center supplies an effective stopping bound. The finiteness theorem used in the last step is classical in characteristic-zero coefficients; we use the formulation in [2, Theorem 1.1].

These are finite algebra parameters. The realization is an explicitly specified quotient of a compact induction from ZD_e. In Section 8, nilpotence exponent one yields an irreducible compact induction from a vertex stabilizer. The article does not assert that every parameter can be converted to a compact inducing pair.

2. Cartan operators and averaging

Choose a maximal F-split torus S and a special vertex x in its reduced apartment. Let K be the full stabilizer of x in G. Thus $Z \subseteq K$ and K/Z is compact. Let $M = C_{\mathbf{G}}(S)(F)$ be the minimal Levi, and choose simple relative roots $\alpha_1, \dots, \alpha_\ell$ and the corresponding positive system. For any standard parabolic $P = LN$, write $\overline{P} = L\overline{N}$ for its opposite with the same Levi.

We use the standard compact open subgroups at x. They can be chosen as a decreasing neighborhood basis D_e, normal in K, with simultaneous Iwahori decompositions for all these parabolics. The required facts are [4, Section I.2, Proposition 7 and Corollaries 8–9]; a formulation including the full reduced-building stabilizer is [5, Theorem 3.11(a)–(d)]. We record the precise properties used below.

Lemma 2.1. *For every $D = D_e$ and every standard parabolic $P = LN$, multiplication gives an Iwahori decomposition*

$$D = (D \cap N)(D \cap L)(D \cap \overline{N}).$$

The subgroup $D \cap M$ is normal in M. An element $a \in M$ whose translation is zero on the roots of L normalizes $D \cap L$. If its coordinates contract the positive root groups in N, then

$$a(D \cap N)a^{-1} \subseteq D \cap N, \qquad a^{-1}(D \cap \overline{N})a \subseteq D \cap \overline{N}.$$

When all these contraction coordinates are strictly positive, the groups $a^{-n}(D \cap N)a^n$ increase to N.

Proof. The cited subgroup construction gives normality and the Iwahori factorizations. In the notation of [4, Section I.2], the minimal-Levi intersection is the good-filtration subgroup H_{e+}, normal in $N_G(S)$, hence normal in M. Conjugation by M shifts the relative root filtrations by its translation coordinates. Zero coordinates leave the Levi filtrations unchanged; positive coordinates give the two inclusions. On every root of N, strict positivity makes the filtration index tend to $-\infty$ after conjugation by a^{-n}. The ordered root-group decomposition of N then proves the exhaustion assertion. These statements use valued root groups, including multipliable roots, and do not require a split group or a tame extension. □

Write $h : M \to \mathbf{R}^\ell$ for the simple-root contraction coordinates. Our sign convention is that $h_i(a) > 0$ contracts the corresponding positive root group. The image $\Lambda = h(M)$ is a lattice, and $\ker h = M \cap K$. Indeed, M acts on the reduced apartment by translations; its full translation lattice is commensurable with the image of $X_*(S)$ under evaluation at a uniformizer. The central translation subspace is spanned by a sublattice, so quotienting it gives a lattice in the reduced apartment. The simple-root forms are rational on this lattice and give coordinates on the apartment; see [5, Lemma 3.3(b)] and its building construction. Put $M^+ = h^{-1}(\mathbf{R}^\ell_{\geq 0})$. Iwasawa and Cartan decomposition give

$$G = PK \quad \text{for every standard } P, \qquad G = KM^+K. \tag{2.1}$$

One may use [5, Theorems 3.13–3.14] and pass from the extended stabilizer to K; the central translation directions disappear in the reduced apartment.

Fix $D = D_e$ and a smooth ω trivial on $Z \cap D$. Let $\mathcal{H}_\omega$ be the convolution algebra of locally constant functions on G, compactly supported modulo Z, satisfying $f(zg) = \omega(z)^{-1} f(g)$. Convolution is integration over G/Z. Let ϵ be the normalized averaging idempotent on ZD with this central character. It acts as the usual D-average e_D on representations with central character ω. Put

$$\mathcal{A} = \epsilon \mathcal{H}_\omega \epsilon, \qquad [g] = \epsilon \delta_g \epsilon \quad (g \in G),$$

where δ_g denotes translation as a convolution multiplier. Thus $[g]$ acts on V^D as $e_D \pi(g)|_{V^D}$.

Lemma 2.2. *For $a, b \in M^+$ and $k_1, k_2 \in K$,*

$$[a][b] = [ab], \qquad [k_1 a k_2] = [k_1][a][k_2]. \tag{2.2}$$

If a strictly contracts the radical N of a standard parabolic LN and normalizes $D \cap L$, then, for $v \in V^D$,

$$e_D \pi(a^n) v = e_{D \cap N} \pi(a^n) v = \pi(a^n) e_{a^{-n}(D \cap N) a^n} v \quad (n \geq 1). \tag{2.3}$$

Proof. Set $N_D = D \cap N$, $L_D = D \cap L$, and $\overline{N}_D = D \cap \overline{N}$. The Iwahori decomposition gives $e_D = e_{N_D} e_{L_D} e_{\overline{N}_D}$. Indeed, $N_D L_D$ is a compact subgroup and $D = (N_D L_D)\overline{N}_D$. The pushforward of normalized Haar measure under this product is normalized Haar measure on D: the first factor acts transitively on $D/\overline{N}_D$. Apply the same argument to $N_D L_D$ to obtain the displayed factorization.

The vector $\pi(a^n)v$ is fixed by L_D and $\overline{N}_D$, by Lemma 2.1. This proves the first equality in (2.3); moving the average across $\pi(a^n)$ proves the second. Apply the same calculation to the minimal parabolic, allowing zero contraction coordinates. Since $aN_D a^{-1} \subseteq N_D \subseteq D$,

$$e_D \pi(a) e_D \pi(b) v = e_D \pi(a) e_{N_D} \pi(b) v = e_D e_{a N_D a^{-1}} \pi(ab) v = e_D \pi(ab) v.$$

This holds in every smooth representation, hence in the faithful regular Hecke action. It proves the first identity of (2.2), without requiring a and b to commute. Normality of D in K proves the second. □

Choose nonzero Hilbert generators $\gamma_1, \dots, \gamma_t$ for the monoid $\Lambda \cap \mathbf{R}^\ell_{\geq 0}$, and choose $b_j \in M^+$ with $h(b_j) = \gamma_j$. This is a finitely generated monoid because the cone is rational polyhedral. Every $a \in M^+$ has an ordered factorization

$$a = m_0 b_1^{n_1} \cdots b_t^{n_t}, \qquad m_0 \in M \cap K, \quad n_j \geq 0. \tag{2.4}$$

To obtain it, express $h(a)$ in the Hilbert generators and take the ordered product; the remaining factor lies in $\ker h$. For $N \geq 1$ put

$$T_j = [b_j], \qquad I_N = \sum_j \mathcal{A} T_j^N \mathcal{A}, \qquad Q_N = \mathcal{A}/I_N. \tag{2.5}$$

In relative semisimple rank zero there are no T_j, and $Q_N = \mathcal{A}$. The quotient is allowed to be zero.

3. A finite calculation of the quotient

The algebra $\mathcal{A}$ has at most one basis vector supported on each ZD double coset. Some cosets may support no vector because of the central character; they are omitted. Each nonzero $[g]$ is a scalar multiple of the corresponding basis vector. Equations (2.2) and (2.4) show that $\mathcal{A}$ is generated by the finite collection

$$\mathcal{S} = \{[k] : k \in K/(ZD)\} \cup \{T_j : 1 \leq j \leq t\}.$$

Choose representatives throughout and discard zero elements. Scalar central factors in products are evaluated using ω.

Let C_N be the finite set of nonzero double-coset basis vectors supported in

$$\bigcup_{0 \leq n_j < N} K b_1^{n_1} \cdots b_t^{n_t} K. \tag{3.1}$$

Write V_N for their span, U_N for the span of all the omitted basis vectors, and $p_N : \mathcal{A} \to V_N$ for the basis projection. Thus $\mathcal{A} = V_N \oplus U_N$ as vector spaces. Cartan factorization gives $U_N \subseteq I_N$. A power T_j^N can nevertheless have retained support if $N\gamma_j$ has another expression using the Hilbert basis. Accordingly,

$$I_N = \text{the two-sided ideal generated by } U_N \text{ and } T_1^N, \dots, T_t^N. \tag{3.2}$$

In particular Q_N is finite-dimensional and

$$\dim Q_N \le [K : ZD]^2 N^t. \tag{3.3}$$

The following finite procedure computes the relations among the retained basis vectors. It does not assume that truncation of convolution is associative. Let $\mathcal{C}_N$ be the support set in (3.1). For each $s \in \mathcal{S}$, only finitely many omitted double cosets meet

$$\operatorname{supp}(s)^{-1}\mathcal{C}_N \ \cup \ \mathcal{C}_N \operatorname{supp}(s)^{-1}.$$

Both sets are compact modulo Z. Compute $p_N(su)$ and $p_N(us)$ for the omitted basis vectors in these finite sets. Let $R_0 \subseteq V_N$ be the span of these boundary vectors together with $p_N(T_1^N), \dots, p_N(T_t^N)$. Starting from R_0, repeatedly adjoin

$$p_N(sR_j) \quad \text{and} \quad p_N(R_j s), \qquad s \in \mathcal{S}.$$

After at most $\dim V_N$ strict increases, the sequence stabilizes at a subspace R_N.

Lemma 3.1. *The preceding construction satisfies*

$$I_N = U_N \oplus R_N, \qquad Q_N = V_N/R_N.$$

Multiplication in Q_N is induced by $v, w \mapsto p_N(vw)$. Thus this finite computation retains every relation contributed by double cosets outside the Cartan box.

Proof. For an omitted basis vector u, the coefficient of a retained coset in su can be nonzero only if its support meets $\operatorname{supp}(s)^{-1}\mathcal{C}_N$. The right-product assertion is the analogous support inclusion. Hence every product of an omitted basis vector with a generator belongs to $U_N + R_0$. By construction, multiplication of R_N by any generator, on either side, belongs to $U_N + R_N$. Since $\mathcal{S}$ generates $\mathcal{A}$, the space $U_N + R_N$ is a two-sided ideal containing U_N and every T_j^N. Equation (3.2) implies $I_N \subseteq U_N + R_N$.

Conversely $U_N \subseteq I_N$. If $u \in U_N$ then $su, us \in I_N$, and subtracting their omitted-basis components shows that $p_N(su), p_N(us) \in I_N$. Also $p_N(T_j^N) \in I_N$ because its omitted component belongs to $U_N \subseteq I_N$. This proves $R_0 \subseteq I_N$. The same argument at every closure step gives $R_N \subseteq I_N$. The sum is direct because $R_N \subseteq V_N$. The asserted multiplication now follows from the quotient of the associative algebra $\mathcal{A}$. □

This is an effective calculation under the usual input assumptions of effective local group arithmetic: finite double-coset decompositions of compact sets, multiplication in their finite quotients, and evaluation of the central character. All boundary products are products of explicitly given compact sets. Their double-coset lists are finite, and the final closure is linear algebra with the displayed dimension bound. An abstract existence theorem for a uniform admissibility bound is not needed.

4. Reconstruction and supercuspidality

For a left Q_N-module L, form the smooth representation

$$\mathcal{V}(L) = \mathcal{H}_\omega \epsilon \otimes_{\mathcal{A}} L.$$

It is generated by its D-fixed vectors, and the unit identity gives $\mathcal{V}(L)^D = L$. In general such induction can have a nonzero subrepresentation with no D-fixed vectors. The next argument rules this out in the present cuspidal situation.

For a parabolic $P = LN$, the unnormalized Jacquet module is the quotient of V by the span of $\pi(n)v - v$, for $n \in N$ and $v \in V$. For irreducible smooth complex representations, vanishing of these quotients for every proper P is equivalent to supercuspidality.

Lemma 4.1. *For every Q_N-module L, all proper Jacquet modules of $\mathcal{V}(L)$ vanish.*

Proof. For each j, let $P_j = M_j N_j$ be the standard parabolic whose Levi roots vanish on γ_j, and put $N_{j,0} = D \cap N_j$. The element b_j normalizes $D \cap M_j$: it shifts none of the root filtrations in this Levi, and $D \cap M$ is normal in M. It strictly contracts N_j and strictly expands its opposite radical. Thus Lemma 2.2 gives, for $v \in \mathcal{V}(L)^D$,

$$T_j^N v = e_D \pi(b_j^N) v = \pi(b_j^N) e_{b_j^{-N} N_{j,0} b_j^N} v.$$

The left side is zero by the definition of Q_N. Hence the image of v in the N_j-coinvariants is zero. Iwasawa decomposition $G = P_j K$ and normality of D in K imply that these images generate the entire Jacquet module: $\mathcal{V}(L)$ is generated by its D-fixed vectors, and K preserves that space. Thus the Jacquet module is zero. The Hilbert basis contains the primitive lattice generator on every extreme ray: a point on such a ray can only be decomposed as a sum of points on the same ray. The parabolics P_j therefore include every maximal standard

parabolic. Conjugation and transitivity of Jacquet functors give the assertion for every proper parabolic. In relative rank zero there are no proper parabolics and the assertion is immediate. □

Lemma 4.2. *An irreducible supercuspidal representation is projective in the category of smooth representations with its actual central character fixed.*

Proof. We recall the coefficient proof of [1, Theorem 5.4.1]. First suppose the central character is unitary. A supercuspidal is unitarizable: integrate the squared absolute value of a nonzero matrix coefficient over G/Z to obtain a positive invariant form. The integral is finite because supercuspidal matrix coefficients have compact support modulo Z. For a unit vector v, with the inner product linear in its first argument, Schur orthogonality gives the idempotent $f(g) = d_\pi \langle \pi(g^{-1})v, v\rangle$ in $\mathcal{H}_\omega$, where d_π is the formal degree. Its left ideal $\mathcal{H}_\omega f$ is isomorphic to the representation. Consequently,

$$\mathrm{Hom}_{\mathcal{H}_\omega}(\mathcal{H}_\omega f, V) \simeq fV.$$

Multiplication by an idempotent is an exact functor, so this module is projective. For general ω, twist by a positive unramified character whose restriction to Z is $|\omega|^{-1}$. Such an extension exists on the real character spaces of the central valuation lattices; see [1, Lemma 5.2.5]. Twisting is an exact equivalence and transfers projectivity back. This is the usual fixed-central-character projectivity argument; it uses no compact-induction assertion. □

Theorem 4.3. *The algebra Q_N is finite-dimensional and semisimple. Its simple modules are in bijection with the irreducible supercuspidal representations π having central character ω, nonzero D-fixed vectors, and*

$$(e_D \pi(b_j)|_{\pi^D})^N = 0 \qquad (1 \le j \le t).$$

The two maps are $L \mapsto \mathcal{V}(L)$ and $\pi \mapsto \pi^D$. Every irreducible supercuspidal with central character ω and $\pi^D \neq 0$ occurs for some N, and one may take $N = \dim \pi^D$.

Proof. First take $L = Q_N$ with its regular left action. The representation $\mathcal{V}(Q_N)$ is generated by the finite-dimensional space Q_N and is cuspidal by the lemma. Every nonzero finitely generated smooth representation has an irreducible quotient: a maximal proper submodule exists because the union of a chain of proper submodules cannot contain a fixed finite generating set. Any irreducible quotient of $\mathcal{V}(Q_N)$ is supercuspidal, by exactness of Jacquet functors. Projectivity splits off that quotient.

The complement is still generated by its D-fixed vectors, because it is the image of a G-equivariant projection of a D-generated representation. Every nonzero quotient of a D-generated representation has nonzero D-fixed vectors. Thus each splitting strictly decreases the dimension of the remaining D-fixed space. After at most $\dim Q_N$ splittings, the complement is zero. This proves that $\mathcal{V}(Q_N)$ is a finite direct sum of irreducible supercuspidals.

For an irreducible smooth π with $\pi^D \neq 0$, its D-fixed space is a simple $\mathcal{A}$-module. Indeed, a nonzero $\mathcal{A}$-submodule generates π under G, and averaging that generated space by e_D recovers exactly the original submodule. Taking D-fixed vectors in the preceding direct sum therefore makes the regular module Q_N a direct sum of simple Q_N-modules. Hence Q_N is semisimple.

For a simple Q_N-module L, the same splitting argument applied to $\mathcal{V}(L)$ gives a finite supercuspidal direct sum. Its D-fixed space is the simple module L, so the sum has exactly one irreducible summand, with multiplicity one. Conversely, if π satisfies the displayed nilpotence conditions, its fixed space is a simple Q_N-module. The natural nonzero map $\mathcal{V}(\pi^D) \to \pi$ is then an isomorphism.

Finally, for a supercuspidal π the Jacquet averaging argument makes each T_j locally nilpotent on π^D: the increasing compact groups $b_j^{-n} N_{j,0} b_j^n$ exhaust N_j, and a vector with zero image in the Jacquet module is killed by averaging over a sufficiently large one of these groups. This space is finite-dimensional by admissibility. A nilpotent operator on a space of dimension d has dth power zero, proving the last assertion. □

The index N bounds nilpotence exponents, not the dimension of a simple module. A simple Q_N-module may have dimension larger than N. The inclusions $I_{N+1} \subseteq I_N$ give surjections $Q_{N+1} \to Q_N$, and the union of their simple spectra gives exactly the supercuspidal representations with these fixed D and ω. The construction does not by itself provide an effective stabilization index for this sequence. Section 5 supplies such an index by a separate finite certificate for the finiteness theorem for Hecke algebras.

5. A certified stopping bound

Write ϵ_e, $\mathcal{A}_{e,\omega}$, and $Q_{e,\omega,N}$ for the preceding idempotent, corner algebra, and quotient formed using $D = D_e$ and the central character ω. We have computed the quotient for each specified N. We now remove the search for a sufficiently large N. The bound will depend on G, the chosen subgroup D_e, and the fixed conventions for local group computations, but not on ω or on a supercuspidal representation. Its existence uses the classical finiteness of a Hecke algebra over its center. We give a finite certificate for this finiteness, so that an effective search for the certificate has a proved stopping condition. This is an application of the finiteness theorem, not a new proof of that theorem.

For this purpose, work first without fixing a central character. Put $\mathcal{H}_D = \mathbf{Q}[D\backslash G/D]$, with Haar measure normalized by $\mathrm{vol}(D) = 1$. Its double-coset basis has rational structure constants, and each product has finite support. The effective input consists of an enumeration of these double cosets, their finite convolution products, and the Cartan and finite quotient operations used above. These are requirements on an encoding of the input, rather than restrictions on residue characteristic, ramification, or the group.

There is a finite, explicitly specified set $\mathcal{S}_D$ of algebra generators. Indeed, choose representatives of $K/(ZD)$, generators and their inverses of the finitely generated abelian group $Z/(Z \cap D)$, and the positive Cartan elements b_j. Take the corresponding compressed operators. An element of K is a product of a central element, one of the chosen representatives, and an element of D. The Cartan decomposition and the positive multiplication identity (2.2), applied before specialization of the central character, then express every double-coset basis vector in the algebra generated by $\mathcal{S}_D$. The finite generation of $Z/(Z \cap D)$ follows from the compactness of the maximal bounded subgroup of the center and the finite rank of its valuation lattice.

We use the following algebraic observation. All elements appearing in a certificate have finite support in the specified algebra basis.

Lemma 5.1. *Let A_0 be a unital rational algebra with an effective countable basis, effective finite products, and a specified finite generating set S. Suppose $A_0 \otimes_{\mathbf{Q}} E$ is finite over its center for some field extension $E/\mathbf{Q}$. There is a terminating search for elements $z_1, \dots, z_r, a_1, \dots, a_m$ in A_0 and polynomials $f_{s,j,k} \in \mathbf{Q}[X_1, \dots, X_r]$ such that*

$$a_1 = 1, \qquad [z_i, s] = 0, \qquad sa_j = \sum_{k=1}^{m} f_{s,j,k}(z_1, \dots, z_r)a_k \quad (s \in S). \tag{5.1}$$

For every such certificate, every finite-dimensional irreducible complex A_0-module V satisfies $(\dim V)^2 \le m$.

Proof. Since S generates A_0, its center is the kernel of the rational linear map

$$A_0 \longrightarrow \bigoplus_{s \in S} A_0, \qquad a \longmapsto ([a, s])_{s \in S}.$$

The set S is finite, so flat base change gives $Z(A_0) \otimes E = Z(A_0 \otimes E)$. Choose finitely many module generators of $A_0 \otimes E$ over this center, and write each as a finite sum of rational algebra elements with coefficients in E. Collect these rational elements as $a_1, \dots, a_m$, adding $a_1 = 1$ if necessary. The module $M_0 = \sum_j Z(A_0)a_j$ satisfies $M_0 \otimes E = A_0 \otimes E$. Faithfulness of extension of scalars implies $M_0 = A_0$.

For each $s \in S$ and each j, express sa_j in these module generators with coefficients in $Z(A_0)$. There are finitely many such coefficients; list them as $z_1, \dots, z_r$. This supplies a rational certificate, with polynomials of degree at most one. It is unnecessary to compute the entire center first.

Each displayed condition is a finite equality in the specified basis. Enumerate all finite lists of finite-support elements with rational coefficients and all finite lists of rational polynomials. Check the identities by exact convolution and rational arithmetic. An enumeration by finite descriptions ensures that every candidate is eventually examined. The existence just proved ensures that the search terminates.

Conversely, a passing certificate suffices on its own. The equations $[z_i, s] = 0$ make every z_i central, since S generates the algebra. The module

$$M = \sum_{k=1}^{m} \mathbf{C}[z_1, \dots, z_r]a_k$$

contains 1 and is stable under left multiplication by every $s \in S$. Induction on word length gives $M = A_0 \otimes \mathbf{C}$. On an irreducible finite-dimensional module V, each z_i acts by a scalar. The image of the algebra in $\mathrm{End}_{\mathbf{C}}(V)$ is therefore spanned by the images of $a_1, \dots, a_m$. Burnside's theorem identifies this image with the full endomorphism algebra. Its dimension is $(\dim V)^2$, which proves the bound. □

For $A_0 = \mathcal{H}_D$, apply [2, Theorem 1.1] with $R = \mathbf{Q}_\ell$, where $\ell \neq p$. Its hypotheses hold for every compact open subgroup over every nonarchimedean local field. Thus $\mathcal{H}_D \otimes \mathbf{Q}_\ell$ is finite over its center, as required by the lemma. The certificate itself is rational; no computation with arbitrary ℓ-adic numbers is needed.

Fix one deterministic enumeration of the certificates, and let m_e be the number of module generators in its first passing certificate for D_e. Define

$$B_e = \max\{1, \lfloor \sqrt{m_e} \rfloor\}.$$

This integer is computed before a central character is supplied. Irreducible smooth complex representations are admissible; see, for example, [3, Proposition 4.10(iv)] for arbitrary local field characteristic. Admissibility and the usual fixed-vector simplicity imply that π^{D_e} is a finite-dimensional simple module of the ordinary Hecke algebra whenever π is irreducible and $\pi^{D_e} \neq 0$. Lemma 5.1 therefore gives

$$\dim \pi^{D_e} \leq B_e. \tag{5.2}$$

Theorem 5.2. *For every smooth central character ω trivial on $Z \cap D_e$, put $Q^{\mathrm{sc}}_{e,\omega} = Q_{e,\omega,B_e}$. Its simple modules give all irreducible supercuspidal representations with central character ω and nonzero D_e-fixed vectors. For every $N \geq B_e$, the canonical surjection*

$$Q_{e,\omega,N} \longrightarrow Q^{\mathrm{sc}}_{e,\omega}$$

is an isomorphism. Computing B_e, forming this finite algebra, and decomposing it is a terminating procedure under the effective input conventions already stated.

Proof. If π is supercuspidal with nonzero D_e-fixed vectors, every positive Cartan operator on π^{D_e} is nilpotent. Its nilpotence index is at most $\dim \pi^{D_e} \leq B_e$. Thus Theorem 4.3 puts every such π in the simple spectrum of $Q^{\mathrm{sc}}_{e,\omega}$.

For $N \geq B_e$, take any simple $Q_{e,\omega,N}$-module L. The same theorem realizes it as π^{D_e} for a supercuspidal π. The dimension bound implies $T_j^{B_e} L = 0$ for every j, so L factors through $Q^{\mathrm{sc}}_{e,\omega}$. The kernel of the displayed surjection therefore annihilates every simple module of $Q_{e,\omega,N}$. That algebra is semisimple, hence the kernel is zero. The certificate search terminates by Lemma 5.1; the boundary calculation terminates by Lemma 3.1. Decomposition of the resulting finite semisimple algebra completes the procedure. The coefficient interface for the specified central character includes exact arithmetic, decidable equality, and algebraic extensions needed to split finitely many polynomials. This is separate from the certificate search, which uses rational arithmetic only. □

An equality between two successive quotients, by itself, is not a valid stopping certificate. For example, let $A = \mathbf{C} \oplus M_r(\mathbf{C})$, let $T = (0, J_r)$ with J_r a nilpotent Jordan block, and take $r \geq 3$. Then $A/(AT^N A) = \mathbf{C}$ for $1 \leq N < r$, whereas $A/(AT^r A) = A$. This finite algebra example concerns the proposed stopping rule, not an assertion that these matrices occur in a reductive-group Hecke algebra. The bound B_e avoids reliance on an unverified plateau.

6. Parameters and the group action

Corollary 6.1. *Fix the vertex, filtration, Cartan representatives, and certificate enumeration. All irreducible supercuspidal representations are parametrized by triples (e, ω, c), where $e \geq 0$, ω is smooth and trivial on $Z \cap D_e$, and c is a primitive central idempotent of the explicitly computed finite algebra $Q^{\mathrm{sc}}_{e,\omega}$. When $e > 0$, require*

$$u_{e-1} c = 0,$$

where u_{e-1} is the averaging idempotent of D_{e-1}, viewed in the D_e-corner and then in $Q^{\mathrm{sc}}_{e,\omega}$. Two such triples give isomorphic representations if and only if they are equal.

Proof. Each primitive central idempotent c cuts out a simple matrix algebra $Q^{\mathrm{sc}}_{e,\omega} c$, with exactly one isomorphism class of simple left modules. Choose a primitive idempotent p in that summand and put $L = Q^{\mathrm{sc}}_{e,\omega} p$. The corresponding representation is

$$\Pi(e, \omega, c) = \mathcal{H}_\omega \epsilon_e \otimes_{\epsilon_e \mathcal{H}_\omega \epsilon_e} L.$$

Different choices of p give isomorphic simple modules, and hence isomorphic representations. The preceding theorem and Theorem 4.3 prove irreducibility, supercuspidality, and exhaustion at the chosen level.

The representation of the matrix summand on L is faithful. Consequently $u_{e-1} c = 0$ is equivalent to $u_{e-1} L = 0$, and hence to the vanishing of $\Pi(e, \omega, c)^{D_{e-1}}$. Since the D_e form a decreasing neighborhood basis, this makes e exactly the least level with nonzero invariants. The central character and this least level are intrinsic. At equal e and ω, distinct primitive central idempotents give nonisomorphic simple modules, proving the asserted

equivalence relation. Conversely, for any supercuspidal π, take its least level e, its central character ω, and the central idempotent supporting the simple module π^{D_e}. These form the required triple. □

6.1. **Finite matrix data.** Fix (e,ω), put $Q = Q^{\mathrm{sc}}_{e,\omega}$, and assume $Q \neq 0$. Choose a basis $q_1 = 1, q_2, \dots, q_n$ and its multiplication constants c^k_{ij}. When $e > 0$, compute $u_{e-1} = \sum_i \eta_i q_i$ in this basis. Explicitly, before passing to Q, this element is

$$u_{e-1} = \frac{1}{[D_{e-1} : D_e]} \sum_{h \in D_{e-1}/D_e} [h].$$

If ω is nontrivial on $Z \cap D_{e-1}$, this average is zero in the central-character algebra. No extra restriction on that intersection is imposed on the data.

Theorem 6.2. *The irreducible supercuspidal representations are parametrized by data*

$$\mathcal{D} = (e, \omega, d; X_1, \dots, X_n),$$

where $1 \le d \le B_e$, the matrices satisfy (1.1)*, and, when $e > 0$, they also satisfy $\sum_i \eta_i X_i = 0$. The associated representation is*

$$\Pi(\mathcal{D}) = \mathcal{H}_\omega \epsilon_e \otimes_{\mathcal{A}_{e,\omega}} \mathbf{C}^d, \qquad q_i v = X_i v. \tag{6.1}$$

Two data are equivalent exactly when their e, ω, and d agree and an invertible matrix S satisfies $SX_i = X_i' S$ for all i. For two valid data with the same e and ω, it is enough to find a nonzero simultaneous intertwiner; this is a finite system of linear equations.

Proof. The multiplication equations and $X_1 = I_d$ define a unital representation of Q on $\mathbf{C}^d$. Its image is precisely the linear span of the X_i, because the q_i form a basis of the algebra. Burnside's theorem says that this image is $M_d(\mathbf{C})$ exactly when the module is simple. Thus the rank condition in (1.1) is exactly the required irreducibility condition.

Theorem 5.2 and Theorem 4.3 now prove that (6.1) is irreducible supercuspidal, and that every supercuspidal at this level is obtained. Its fixed-vector space is $\mathbf{C}^d$, so (5.2) gives $d \le B_e$. The equation $\sum_i \eta_i X_i = 0$ says that the preceding level has no fixed vectors. It therefore enforces the least level, as in Corollary 6.1. Isomorphisms of the simple modules are precisely simultaneous changes of basis. A nonzero intertwiner between simple modules has zero kernel and full image, hence is an isomorphism. This proves the last assertion and exhaustion without repetitions across levels. □

6.2. **The representation by generators and relations.** We spell out the group representation in (6.1). This also specifies an action for every element of G, rather than only for the finite Hecke generators. Put $D = D_e$ and $L = \mathbf{C}^d$. Choose elements $s_1, \dots, s_r \in G$ such that $a_i = \epsilon_e \delta_{s_i} \epsilon_e$ generate $\mathcal{A}_{e,\omega}$; one may use the representatives in $K/(ZD)$ and the b_j. Let A_i be the matrix of the image of a_i on L, obtained from the matrices X_k and the computed coordinates of that image in Q.

Start with symbols $[g, v]$, for $g \in G$ and $v \in L$, linear in v, and form their finite linear combinations. Impose

$$[gd, v] = [g, v], \qquad [gz, v] = \omega(z)[g, v] \quad (d \in D,\ z \in Z). \tag{6.2}$$

This space is $c\mathrm{Ind}^G_{ZD}(\omega \otimes L)$, where zd acts on L as the scalar $\omega(z)$. Choose representatives $d_{i,1}, \dots, d_{i,m_i}$ of $D/(D \cap s_i D s_i^{-1})$, with $m_i = [D : D \cap s_i D s_i^{-1}]$. Impose in addition

$$\frac{1}{m_i} \sum_{j=1}^{m_i} [g d_{i,j} s_i, v] = [g, A_i v] \quad (g \in G,\ v \in L,\ 1 \le i \le r). \tag{6.3}$$

The group action on the resulting quotient is

$$h[g, v] = [hg, v] \quad (h \in G). \tag{6.4}$$

All vectors are finite sums; there is no completion.

Proposition 6.3. *The quotient defined by* (6.2) *and* (6.3)*, with action* (6.4)*, is canonically isomorphic to* (6.1)*. For every valid datum it is nonzero, irreducible, and supercuspidal. The map $v \mapsto [1, v]$ identifies L with its D-fixed space.*

Proof. Write $u_g = \delta_g * \epsilon_e \in \mathcal{H}_\omega \epsilon_e$. These elements satisfy $u_{gd} = u_g$ and $u_{gz} = \omega(z) u_g$. They form the usual basis with central-character relations for compact induction. Normalized averaging gives

$$u_g * a_i = \frac{1}{m_i} \sum_{j=1}^{m_i} u_{g d_{i,j} s_i}.$$

Thus (6.3) imposes exactly the balancing relation $(u * a_i) \otimes v = u \otimes A_i v$. Relations for the finite generating set imply balancing for a word in the generators by induction on its length, and then for every element of $\mathcal{A}_{e,\omega}$ by linearity. The quotient is consequently the tensor product in (6.1).

Averaging this tensor product by ϵ_e gives $\mathcal{A}_{e,\omega} \otimes_{\mathcal{A}_{e,\omega}} L = L$, with the stated map. Smoothness also follows directly: a finite sum of symbols is fixed by an intersection of finitely many conjugates of D. The remaining claims follow from Theorem 4.3. □

7. The computation and its input

The algebraic assertions above concern every connected reductive G over a nonarchimedean local field and every smooth complex central character. For a literal algorithm, we specify the following effective input.

(i) The chosen subgroups, Cartan lattice and Hilbert generators are available. Ordinary D double cosets can be enumerated and their finite convolution products can be computed exactly. Finite double-coset lists for the compact sets occurring in the boundary calculation are available, with the corresponding central factors when working modulo Z.

(ii) The prescribed central character can be evaluated on the central factors in these operations. The coefficient field containing these values has exact arithmetic and decidable equality, with effective algebraic extensions and factorization sufficient to split the finite algebras that occur.

The character is part of the input. In particular, arbitrary unspecified complex numbers are not being treated as computable numbers, and the procedure does not enumerate all complex central characters.

For a specified (e, ω), the procedure is as follows.

(i) Search the rational certificates of Lemma 5.1 for the ordinary Hecke algebra at D_e. The first passing certificate gives m_e and B_e.

(ii) Put $N = B_e$. List the retained double cosets in (3.1) and compute the boundary relations and the subspace R_N of Lemma 3.1.

(iii) Form the multiplication table of $Q = V_N / R_N$. If $Q = 0$, return the empty list. Otherwise decompose the finite semisimple algebra, choose a simple module for each matrix factor, and compute the matrices of its basis.

(iv) For parameters of least level, when $e > 0$, discard the modules on which u_{e-1} acts nontrivially. Retain one module per matrix factor and use (6.2)–(6.4) for the associated group representation.

The certificate search terminates by finiteness over the center. The boundary closure makes at most $\dim V_N$ strict increases, and the remaining operations are finite algebra calculations. This proves the algorithmic assertion of Theorem 1.1; the reconstruction and exhaustion assertions follow from Theorem 5.2 and Theorem 6.2. No complexity bound is asserted for the exhaustive certificate search.

8. Compact induction and examples

Proposition 8.1. *Every simple Q_1-module W is an irreducible representation of K, trivial on D and with central character ω, and its associated supercuspidal is*

$$\mathcal{V}(W) \simeq \text{c-Ind}_K^G W.$$

Consequently, the supercuspidals with nonzero D-fixed vectors on which every T_j acts by zero are classified by these irreducible compact inducing data. This assertion holds for every connected reductive group and every residue characteristic.

Proof. For $N = 1$, the support set in (3.1) is K. Thus Q_1 is a quotient of the finite twisted group algebra of $K/(ZD)$ with central character ω. Its simple module W is irreducible as a K-representation. By Theorem 4.3, the associated $\pi = \mathcal{V}(W)$ is irreducible supercuspidal and $\pi^D = W$. Moreover,

$$e_D \pi(g)|_W = 0 \qquad (g \notin K),$$

because every omitted double-coset basis vector is zero in Q_1.

Twist by a positive unramified character to make the central character unitary. This character is trivial on the compact group D, so the vanishing assertion and the identification of D-fixed vectors are unchanged. Equip π with an invariant positive definite Hermitian form. The operator e_D is the orthogonal projection onto W. If $gK \neq hK$, then for $v, w \in W$ one has

$$\langle \pi(g)v, \pi(h)w \rangle = \langle v, e_D \pi(g^{-1}h)w \rangle = 0.$$

Hence the spaces $\pi(g)W$, indexed by G/K, form an algebraic direct sum. They span π because W generates it. The natural map from $\mathrm{c\text{-}Ind}_K^G W$ to π is therefore both injective and surjective. Twisting back proves the assertion for the original central character. □

8.1. **Groups compact modulo their center.** If $\mathbf{G}$ has relative semisimple rank zero, its reduced building is a point, $K = G$, and there are no positive Cartan operators. The algebra $Q_N = \mathcal{A}$ is the finite twisted group algebra of $G/(ZD)$ determined by ω. Its simple modules are already representations of G, trivial on D. There are no proper parabolic subgroups, so they are all supercuspidal. In particular, if $\mathbf{G}$ is a torus, then $Z = G$ and $Q_N = \mathbf{C}$ for each admissible ω. The sole simple module reconstructs the one-dimensional representation ω; the least-level condition records the first D_e contained in its kernel.

8.2. **A concrete sector in residue characteristic two.** Let F have residue field $\mathbf{F}_2$, let $G = \mathrm{PGL}_2(F)$, and take $K = \mathrm{PGL}_2(\mathcal{O}_F)$ and $D = \ker(K \to \mathrm{PGL}_2(\mathbf{F}_2))$. Put $b = \mathrm{diag}(\varpi, 1)$, viewed in G. Then $K/D \simeq S_3$, and there is one positive Cartan generator $T = [b]$.

Proposition 8.2. *For these choices, $Q_1 = \mathcal{A}/(\mathcal{A}T\mathcal{A}) \simeq \mathbf{C}$. Its simple module is the sign character of S_3, inflated to K, and its associated representation is $\mathrm{c\text{-}Ind}_K^G \mathrm{sgn}$. Thus the matrix data in this sector are scalar signs on K/D and the scalar zero on T.*

Proof. The algebra Q_1 is a semisimple quotient of $\mathbf{C}[S_3]$. If W is one of its simple modules, Proposition 8.1 says that $\mathrm{c\text{-}Ind}_K^G W$ is supercuspidal. Let N be a maximal unipotent subgroup with integral subgroup $U = K \cap N$. Its orbit through the identity coset in G/K contributes the summand $\mathrm{c\text{-}Ind}_U^N W$ to the restriction of this compact induction to N. Taking N-coinvariants of that summand gives W_U. Since the whole Jacquet module is zero, $W_U = 0$. The image of U in S_3 is generated by a transposition. Among the trivial, sign, and two-dimensional standard representations of S_3, only sign has zero invariants under that subgroup. In characteristic zero invariants and coinvariants for a finite group have the same dimension. Thus sign is the only possible simple module of Q_1.

It remains to show that it occurs. Write $a_n = \mathrm{diag}(\varpi^n, 1)$ for $n \geq 1$, and let U^+ and U^- be the upper and lower integral unipotent subgroups. Direct multiplication gives

$$a_n^{-1} U^- a_n \subseteq D, \qquad a_n U^+ a_n^{-1} \subseteq D.$$

The sign character is nontrivial on both U^+ and U^-. The first inclusion shows that no a_n intertwines it: on $U^- \subseteq K \cap a_n K a_n^{-1}$, one of the two characters is nontrivial and the other is trivial. Cartan decomposition gives the same conclusion for every element outside K.

For completeness, the sign isotypic space under K in $\Pi = \mathrm{c\text{-}Ind}_K^G \mathrm{sgn}$ is therefore exactly its canonical one-dimensional space supported on K, by the compact Mackey decomposition. Any nonzero subrepresentation contains a vector whose translate has nonzero value on that coset. Projecting to the sign isotypic space gives a nonzero vector in the canonical copy, which generates Π. Hence Π is irreducible. The invariant unitary form on compact induction has compactly supported matrix coefficients for finite-support vectors; admissibility and irreducibility then imply supercuspidality.

The second inclusion kills every D-fixed Mackey term away from K: $U^+ \subseteq K \cap a_n^{-1} D a_n$, whereas sign has no U^+-invariants. Normality of D in K handles the compact factors. Consequently $\Pi^D = \mathrm{sgn}$. Averaging a translate supported outside K cannot create support on K; since every D-fixed vector is supported on K, this average is zero. Thus the compressed action outside K vanishes, and Π^D is a Q_1-module. A finite-dimensional semisimple complex algebra with exactly one simple module, of dimension one, is $\mathbf{C}$, proving the claim. □

This example computes the exponent-one sector. For a general group or level, the stopping bound B_e can be larger than one. The full finite algebra $Q_{e,\omega}^{\mathrm{sc}}$ then still parametrizes the supercuspidals, but the preceding compact-induction argument does not supply an inducing subgroup for every simple module.

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